

Programming Lab 3 — Branching

Case Problem Labs

In this lab, you will continue working with your assigned Case Problem business. *You can find your assigned case problem group by clicking the Groups icon in the main Canvas navigation pane.*

You will create a Python program that uses branching statements (if, elif, and else) to make decisions based on user input and business rules.

This lab reinforces:

- branching statements
- Boolean expressions
- comparison operators
- logical operators
- nested decisions
- calculations
- formatted output
- real-world business logic

Important Note

Programming labs often require multiple attempts and revisions—this is a normal and expected part of learning to program.

★ Challenge Extensions (Optional)

Challenge Extensions are **optional** enhancements for students who would like additional practice or a more advanced programming challenge. These extensions allow you to expand your program with additional features, business rules, or calculations beyond the core lab requirements.

Completing challenge extensions may strengthen rubric categories such as problem-solving, creativity, and program complexity.

The Barking Lot

Fountain View Hall

Cost Is No Object

The Barking Lot

Lab 3: Boarding Eligibility & Pricing System

Overview

The Barking Lot needs a program that determines whether a dog can be accepted for boarding and calculates the total boarding cost based on business rules and optional services.

Program Requirements

Prompt the user for:

- dog name
- vaccination status (yes/no)
- aggressive behavior history (yes/no)
- number of boarding days
- grooming package (yes/no)
- VIP playtime package (yes/no)

Business Rules

Eligibility Rules

- Dogs without vaccinations are automatically denied boarding
- Dogs with an aggressive behavior history require manager approval

Pricing Rules

- Base daily boarding rate
- Boarding stays longer than 10 days receive a discount
- Grooming package adds an additional fee
- VIP playtime package adds an additional fee

Assumptions & Pricing

Not all pricing values and business rules are fully defined. You may make reasonable assumptions for rates, fees, discounts, and surcharges as needed to complete your program. Clearly document any assumptions in your code using comments.

Program Must

- Use branching statements (if, elif, and else)
- Use logical operators
- Calculate total boarding cost
- Display different messages depending on eligibility
- Display a formatted boarding summary
- Include at least 3 comments

Example Output

Boarding Summary

Pet Name: Bella

Vaccination Status: Approved

Manager Approval Required: No

Boarding Days: 12

Long Stay Discount Applied: Yes

Grooming Package: Yes

VIP Playtime: Yes

Final Total: \$412.50

★ Challenge Extension (Optional)

Add:

- multiple dog discounts
- holiday pricing
- medication administration fee
- late pickup fee

Submission Requirements

Submit:

- .py file
 - Test cases
 - Code walk-through video (3–4 minutes)
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Fountain View Hall

Lab 3: Event Booking Approval System

Overview

Fountain View Hall needs a program to evaluate event bookings and calculate pricing based on event size, catering options, and booking requirements.

Program Requirements

Prompt the user for:

- event type
 - guest count
 - weekend booking (yes/no)
 - catering package (standard/premium)
 - deposit paid (yes/no)
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Business Rules

Approval Rules

- Events with over 300 guests require manager approval
- Events cannot be booked without a deposit

Pricing Rules

- Weekend events include a surcharge
- Premium catering adds an additional fee
- Large events receive a discount
- Weddings automatically include a cleanup fee

Assumptions & Pricing

Not all pricing values and business rules are fully defined. You may make reasonable assumptions for rates, fees, discounts, and surcharges as needed to complete your program. Clearly document any assumptions in your code using comments.

Program Must

- Use if, elif, and else statements
 - Use comparison and logical operators
 - Perform calculations using branching decisions
 - Display approval or denial messages
 - Display a formatted booking summary
 - Include at least 3 comments
-

Example Output

Event Booking Summary

Event Type: Wedding
Guests: 250
Weekend Event: Yes
Premium Catering: Yes
Deposit Received: Yes
Large Event Discount Applied: Yes
Cleanup Fee Applied: Yes
Estimated Total: \$5,850.00

★ **Challenge Extension (Optional)**

Add:

- decoration packages
- security fees
- holiday surcharges
- multiple event rooms

Submission Requirements

Submit:

- .py file
 - Test cases
 - Code walk-through video (3–4 minutes)
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Cost Is No Object

Lab 3: Rental Discount Calculator

Overview

Cost Is Not Object wants a program that determines rental discounts based on the number of rental days and customer status.

Program Requirements

Prompt the user for:

- customer name
- customer age
- rental days
- daily rental rate
- luxury vehicle request (yes/no)
- premium membership (yes/no)

Business Rules

Eligibility Rules

- Drivers must be at least 18 years old
- Luxury vehicles require drivers to be at least 25 years old

Pricing Rules

- Premium members receive a discount
- Rentals longer than 7 days receive a discount
- Luxury vehicles include a surcharge
- Drivers under 25 receive a young-driver fee

Assumptions & Pricing

Not all pricing values and business rules are fully defined. You may make reasonable assumptions for rates, fees, discounts, and surcharges as needed to complete your program. Clearly document any assumptions in your code using comments.

Program Must

- Use branching statements (if, elif, and else)
 - Use logical operators
 - Perform calculations using branching decisions
 - Display approval or denial messages
 - Display a formatted rental summary
 - Include at least 3 comments
-

Example Output

Rental Summary

Customer: Alex

Rental Approved: Yes

Luxury Vehicle: Yes

Rental Days: 8

Premium Discount Applied: Yes

Young Driver Fee Applied: No

Final Total: \$845.50

★ Challenge Extension (Optional)

Add:

- insurance package options
- mileage fees
- fuel charges
- tax calculations

Submission Requirements

Submit:

- .py file
 - Test cases
 - Code walk-through video (3–4 minutes)
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